

NAND Implementation

By Hardware / software codesign using Zynq processor

Design Report

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Introduction

To implement the NAND gate by designing an AND gate in hardware and using a c program running on the Zynq processor on the ZedBoard Zynq Evaluation and Development Kit (xc7z020clg484-1) to perform the inversion of the AND gate output.

AND gate code

```
module and_gate(  
    input a,b,  
    output c);  
    assign c = a&b;  
endmodule
```

Figure 1: AND code

Now we need to create a block diagram design and add to it the AND RTL , the Zynq processor then run block automation to connect it with the external DDR memory and then we add two AXI GPIOs and configure one to accept 1 bit input from the output of the OR gate and the other to output 1 bit from the processor, and run connection automation to connect the AXI ports.

Final Hardware System

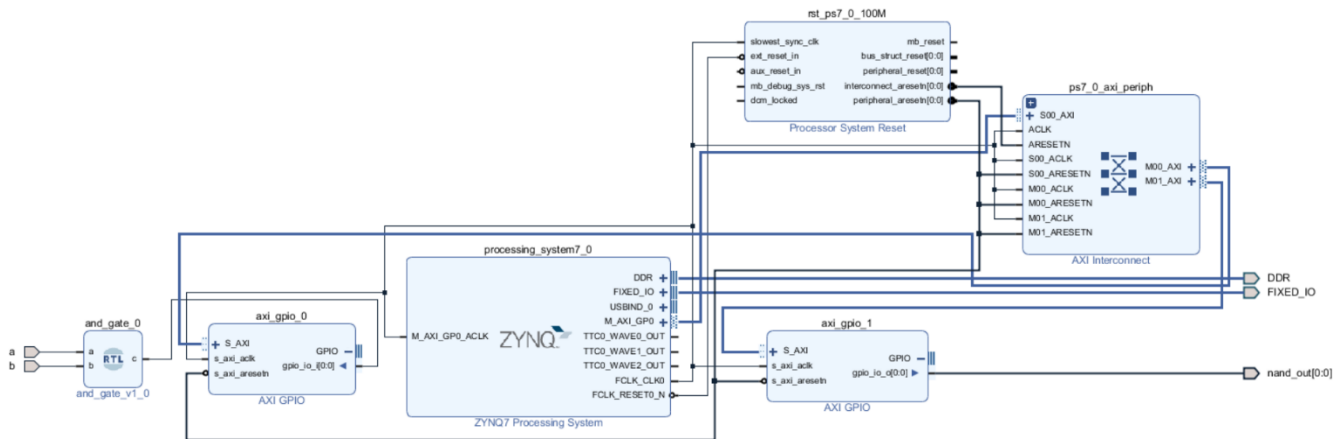


Figure 2 : Final Hardware System

Design Validation

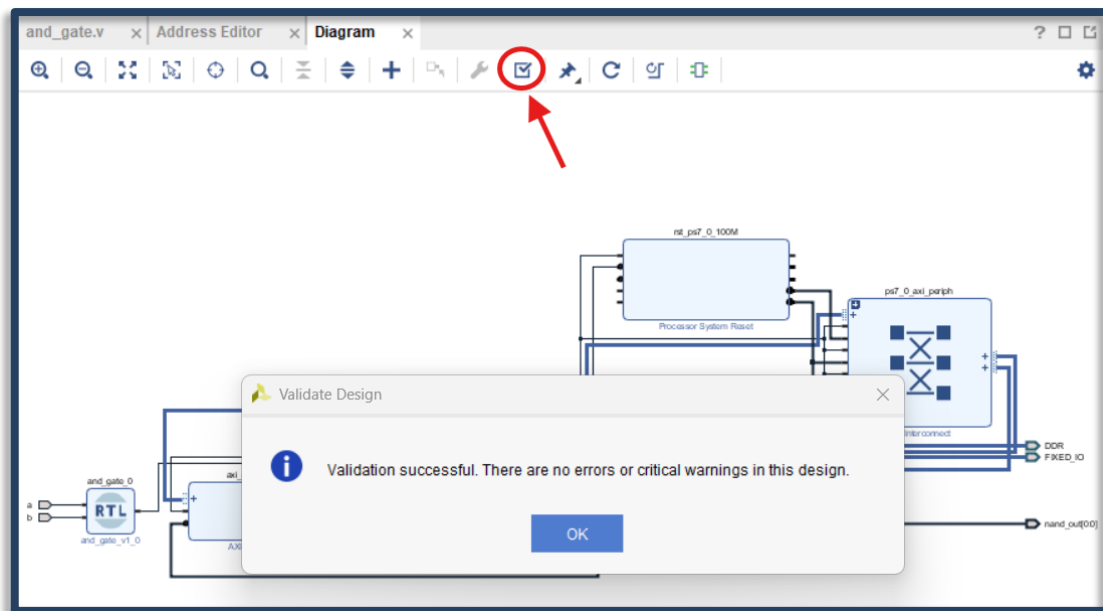


Figure 3 : Design Validation

Now after creating HDL wrapper for the block diagram and making sure it's the top module we need to create a constraint file.

```
# -----  
# User LEDs - Bank 33  
# -----  
set_property PACKAGE_PIN T22 [get_ports {nand_out}]; # "LDO"  
  
## -----  
## User DIP Switches - Bank 35  
## -----  
set_property PACKAGE_PIN F22 [get_ports {a}]; # "SW0"  
set_property PACKAGE_PIN G22 [get_ports {b}]; # "SW1"  
  
# Note that the bank voltage for IO Bank 33 is fixed to 3.3V on SedBoard.  
set_property IOSTANDARD LVCMOS33 [get_ports -of_objects [get_iobanks 33]];  
  
# Set the bank voltage for IO Bank 34 to 1.8V by default.  
# set_property IOSTANDARD LVCMOS33 [get_ports -of_objects [get_iobanks 34]];  
# set_property IOSTANDARD LVCMOS25 [get_ports -of_objects [get_iobanks 34]];  
# set_property IOSTANDARD LVCMOS18 [get_ports -of_objects [get_iobanks 34]];  
  
# Set the bank voltage for IO Bank 35 to 1.8V by default.  
# set_property IOSTANDARD LVCMOS33 [get_ports -of_objects [get_iobanks 35]];  
# set_property IOSTANDARD LVCMOS25 [get_ports -of_objects [get_iobanks 35]];  
set_property IOSTANDARD LVCMOS18 [get_ports -of_objects [get_iobanks 35]];  
  
# Note that the bank voltage for IO Bank 13 is fixed to 3.3V on SedBoard.  
# set_property IOSTANDARD LVCMOS33 [get_ports -of_objects [get_iobanks 13]];
```

Figure 4 : Constraint File

Now we run the full flow till the bitstream generation:

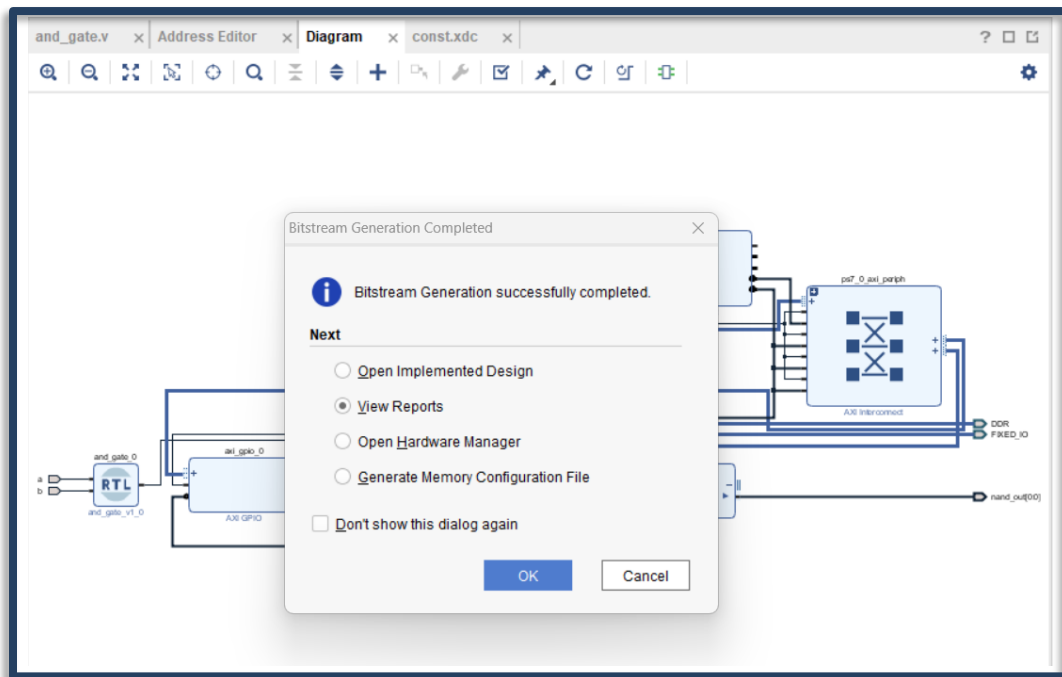


Figure 5 : Bitstream Generation

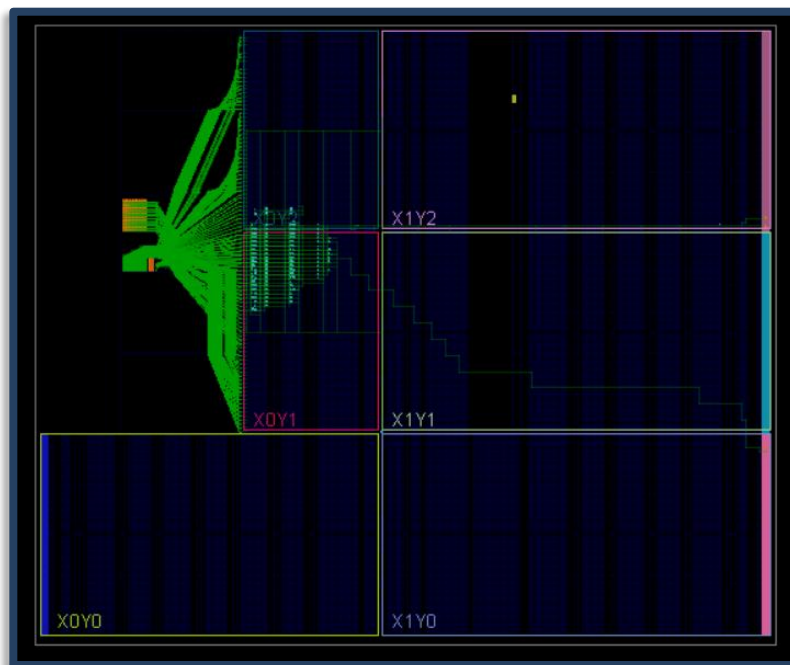
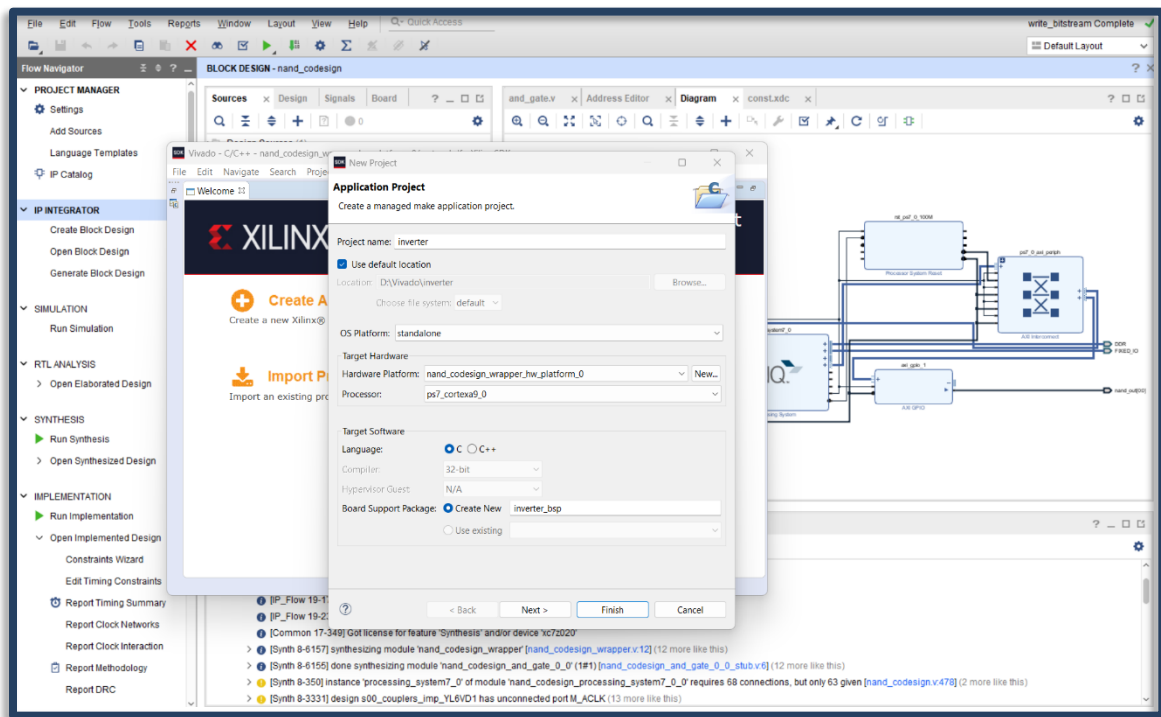
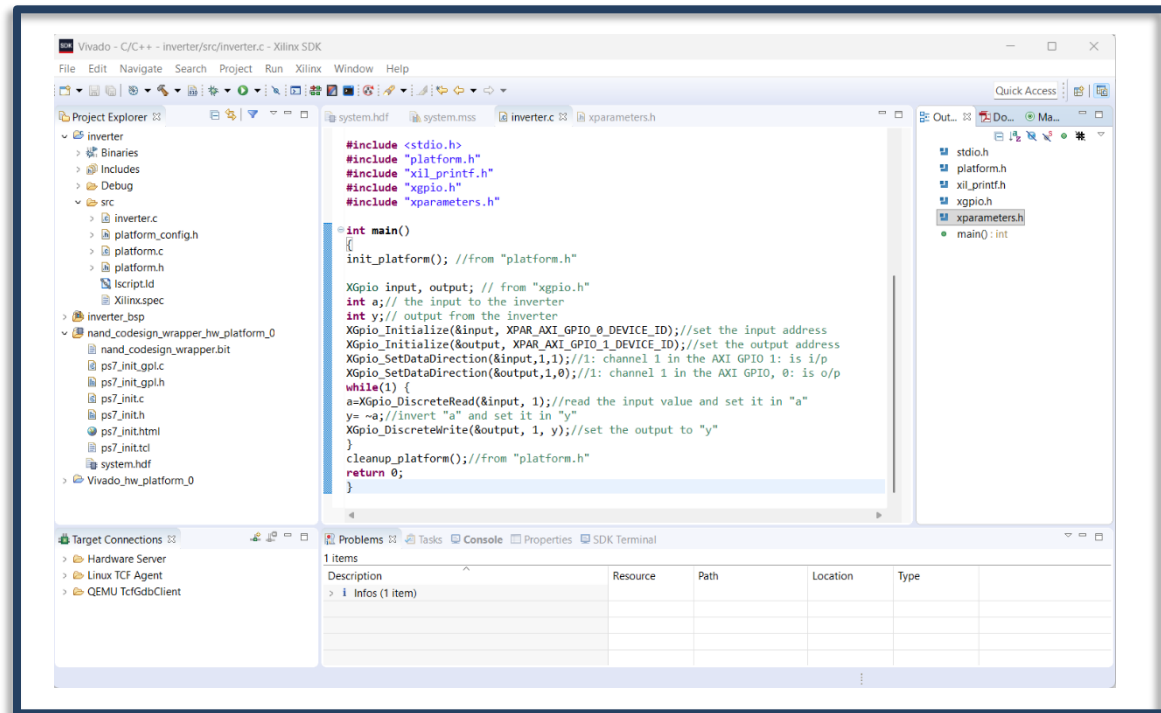


Figure 6 : implemented design

After exporting the hardware (including the bitstream file) we launch SDK and create a new application project



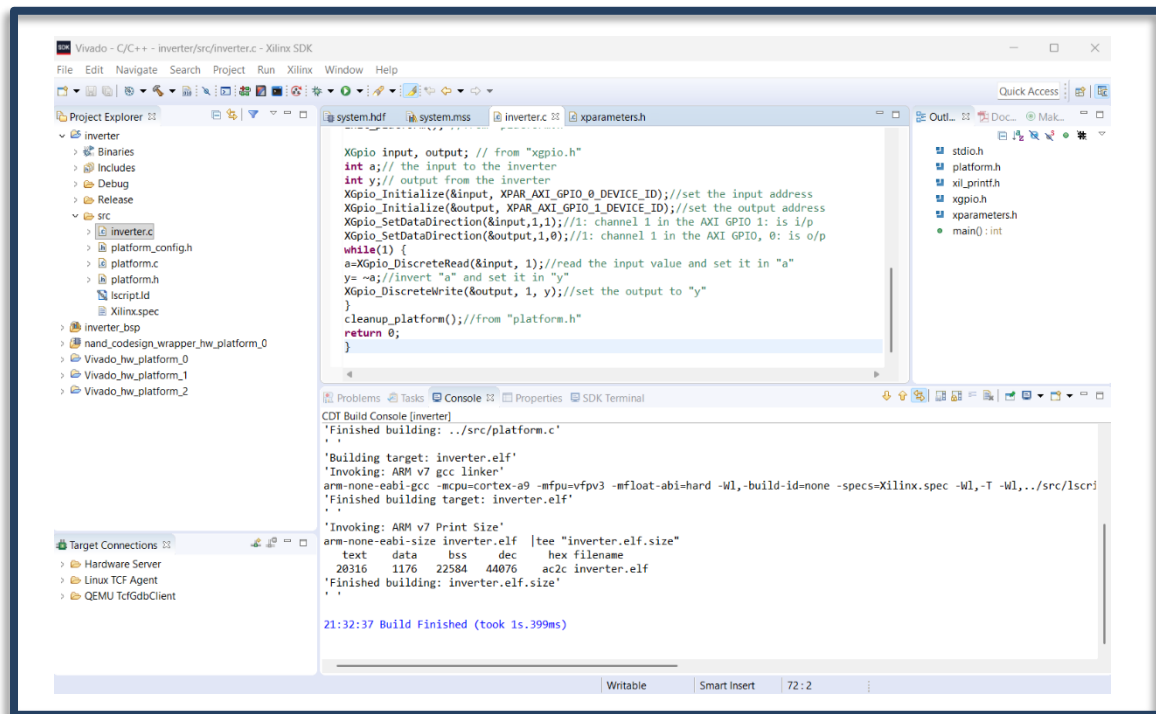
By using one of the available templates and edit it to perform the inverting



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Building the project



The last step is to program the FPGA:

